The game of Euclid (13.34)

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Two players A and B alternately take chips from two piles with a and b chips, respectively. Initially $a > b$. A move consists in taking a multiple of the other pile from a pile. The winner is the one who takes the last chip in one of the piles. Show that

(a) If $a > 2b$, then the first player A can force a win.

(b) For what α can A force a win, if initially $a > \alpha b$. (This game of Euclid is due to Cole and Davie, *Math. Gaz.* LIII, 354–7 (1969).)